

Day-3-Subqueries, Set Operators & CTEs (Assignment-2)

/*1. Write a query to display the average monthly ticket cost for each flight in ABC Airlines. The query should display the Flight_Id, From_location, To_Location, Month Name as "Month_Name" and average price as "Average_Price". Display the records sorted in ascending*/

```
SELECT afd.Flight_ID,af.From_Location,af.To_Location,  
       DATENAME(MONTH,afd.Flight_Departure_Date) AS Month_Name,AVG(afd.Price) AS  
Average_Price  
FROM air_flight_details afd  
JOIN air_flight af  
ON af.Flight_ID=afd.Flight_ID  
GROUP BY  
afd.Flight_ID,af.From_Location,af.To_Location,DATENAME(MONTH,afd.Flight_Departure  
_Date),  
       MONTH(afd.Flight_Departure_Date)  
ORDER BY afd.Flight_ID,MONTH(afd.Flight_Departure_Date);
```

Output:

	Flight_ID	From_Location	To_Location	Month_Name	Average_Price
1	F101	Chennai	Hyderabad	January	4500.000000
2	F101	Chennai	Hyderabad	April	5000.000000
3	F102	Hyderabad	Delhi	February	7200.000000
4	F103	Chennai	Hyderabad	March	4600.000000
5	F103	Chennai	Hyderabad	April	5100.000000
6	F104	Bangalore	Mumbai	February	7000.000000
7	F105	Delhi	Chennai	January	6500.000000
8	F201	Mumbai	Hyderabad	February	6000.000000
9	F202	Hyderabad	Chennai	March	5800.000000
10	F205	Chennai	Hyderabad	April	5300.000000

/*2. Write a query to display the customer(s) who has/have booked least number of tickets in ABC Airlines. The Query should display profile_id, customer's first_name, Address and Number of tickets booked as "No_of_Tickets". Display the records sorted in ascending order based on customer's first name.*/

```
SELECT app.Profile_ID, app.First_Name, app.Address, COUNT(ati.Ticket_ID) AS
No_Of_Tickets
FROM air_passenger_profile app
JOIN air_ticket_info ati
ON app.Profile_ID=ati.Profile_ID
GROUP BY app.Profile_ID, app.First_Name, app.Address
HAVING COUNT(ati.TICKET_ID)=(
    SELECT MIN(ticket_count)
    FROM(
        SELECT COUNT(*) AS ticket_count
        FROM air_ticket_info
        GROUP BY PROFILE_ID
    )x
)
ORDER BY app.First_Name;
```

Output:

Results		Messages		
	Profile_ID	First_Name	Address	No_Of_Tickets
1	P106	Karan	Mumbai	1
2	P105	Nisha	Chennai	1
3	P107	Rahul	Chennai	1
4	P102	Sita	Hyderabad	1

/*3. Write a query to display the number of flight services between locations in a month. The Query should display From_Location, To_Location, Month as "Month_Name" and number of flight services as "No_of_Services". Hint: The Number of Services can be calculated from the number of scheduled departure dates of a flight. The records should be displayed in ascending order based on From_Location and then by To_Location and then by month name.*/

```
SELECT af.From_Location,af.To_Location,DATENAME(MONTH,afd.Flight_Departure_Date)
AS Month_Name,
COUNT(af.Flight_ID) AS No_Of_Services
FROM air_flight af
JOIN air_flight_details afd
ON afd.Flight_ID=af.Flight_ID
GROUP BY
af.From_Location,af.To_Location,DATENAME(MONTH,afd.Flight_Departure_Date),
MONTH(afd.Flight_Departure_Date)
ORDER BY af.From_Location,af.To_Location,MONTH(afd.Flight_Departure_Date);
```

Output:

	From_Location	To_Location	Month_Name	No_Of_Services
1	Bangalore	Mumbai	February	1
2	Chennai	Hyderabad	January	1
3	Chennai	Hyderabad	March	1
4	Chennai	Hyderabad	April	3
5	Delhi	Chennai	January	1
6	Hyderabad	Chennai	March	1
7	Hyderabad	Delhi	February	1
8	Mumbai	Hyderabad	February	1

/*4. Write a query to display the customer(s) who has/have booked maximum number of tickets in ABC Airlines. The Query should display profile_id, customer's first_name, Address and Number of tickets booked as "No_of_Tickets". Display the records in ascending order based on customer's first name.*/

```
SELECT app.Profile_ID,app.First_Name,app.Address, COUNT(ati.Ticket_ID) AS
No_Of_Tickets
FROM air_passenger_profile app
JOIN air_ticket_info ati
ON app.Profile_ID=ati.Profile_ID
GROUP BY app.Profile_ID,app.First_Name,app.Address
HAVING COUNT(ati.TICKET_ID)=(
    SELECT MAX(ticket_count)
    FROM(
        SELECT COUNT(*) AS ticket_count
        FROM air_ticket_info
        GROUP BY PROFILE_ID
    )x
)
ORDER BY app.First_Name;
```

Output:

Results		Messages		
	Profile_ID	First_Name	Address	No_Of_Tickets
1	P103	Arjun	Bangalore	3
2	P104	Isha	Delhi	3
3	P101	Ravi	Chennai	3

/*5. Write a query to display the number of tickets booked from Chennai to Hyderabad. The Query should display passenger profile_id,first_name,last_name, Flight_Id , Departure_Date and number of tickets booked as “No_of_Tickets”. Display the records sorted in ascending order based on profile id and then by flight id and then by departure date.*/

```
SELECT app.Profile_ID,app.First_Name,app.Last_Name,ati.Flight_ID,
       ati.Flight_Departure_Date,COUNT(ati.Ticket_ID) AS No_Of_Tickets
FROM air_passenger_profile app
JOIN air_ticket_info ati
ON ati.Profile_ID=app.Profile_ID
JOIN air_flight af
ON af.Flight_ID=ati.Flight_ID
WHERE af.FROM_LOCATION = 'Chennai' AND af.TO_LOCATION = 'Hyderabad'
GROUP BY app.Profile_ID,app.First_Name,app.Last_Name,ati.Flight_ID,
         ati.Flight_Departure_Date
ORDER BY app.Profile_ID,ati.Flight_ID;
```

Output:

	Profile_ID	First_Name	Last_Name	Flight_ID	Flight_Departure_Date	No_Of_Tickets
1	P101	Ravi	Kumar	F101	2025-01-10	1
2	P101	Ravi	Kumar	F101	2025-04-05	1
3	P102	Sita	Rao	F103	2025-03-20	1
4	P103	Arjun	Mehta	F103	2025-04-25	1
5	P104	Isha	Sharma	F205	2025-04-12	1
6	P105	Nisha	Patel	F205	2025-04-12	1
7	P107	Rahul	Singh	F101	2025-01-10	1

/*6. Write a query to display flight id,from location, to location and ticket price of flights whose departure is in the month of april.*/

```
SELECT af.Flight_ID,af.From_Location,af.To_Location,afd.Price
      FROM air_flight af
JOIN air_flight_details afd
ON afd.Flight_ID=af.Flight_ID
WHERE MONTH(afd.Flight_Departure_Date)=4;
```

Output:

Results		Messages		
	Flight_ID	From_Location	To_Location	Price
1	F101	Chennai	Hyderabad	5000.00
2	F103	Chennai	Hyderabad	5100.00
3	F205	Chennai	Hyderabad	5300.00

/*7. Write a query to display the average cost of the tickets in each flight on all scheduled dates. The query should display flight_id, from_location, to_location and Average price as "Price". Display the records sorted in ascending order based on flight id and then by from_location and then by to_location.*/

```
SELECT af.Flight_ID,af.From_Location,af.To_Location,AVG(afd.Price) AS Price
FROM air_flight af
JOIN air_flight_details afd
ON af.Flight_ID=afd.Flight_ID
GROUP BY af.Flight_ID,af.From_Location,af.To_Location
ORDER BY af.Flight_ID,af.From_Location,af.To_Location;
```

Output:

Results		Messages		
	Flight_ID	From_Location	To_Location	Price
1	F101	Chennai	Hyderabad	4750.000000
2	F102	Hyderabad	Delhi	7200.000000
3	F103	Chennai	Hyderabad	4850.000000
4	F104	Bangalore	Mumbai	7000.000000
5	F105	Delhi	Chennai	6500.000000
6	F201	Mumbai	Hyderabad	6000.000000
7	F202	Hyderabad	Chennai	5800.000000
8	F205	Chennai	Hyderabad	5300.000000

/*8. Write a query to display the customers who have booked tickets from Chennai to Hyderabad. The query should display profile_id, customer_name (combine first_name & last_name with comma in b/w), address of the customer. Give an alias to the name as customer_name. Hint: Query should fetch unique customers irrespective of multiple tickets booked. Display the records sorted in ascending order based on profile id. */

```
SELECT DISTINCT app.Profile_ID, CONCAT(app.First_Name, ', ', app.Last_Name) AS  
Customer_Name, app.Address  
FROM air_passenger_profile app  
JOIN air_ticket_info ati  
ON ati.Profile_ID=app.Profile_ID  
JOIN air_flight af  
ON af.Flight_ID=ati.Flight_ID  
WHERE af.From_Location='Chennai' AND af.To_Location='Hyderabad'  
ORDER BY app.Profile_ID;
```

Output:

	Profile_ID	Customer_Name	Address
1	P101	Ravi,Kumar	Chennai
2	P102	Sita,Rao	Hyderabad
3	P103	Arjun,Mehta	Bangalore
4	P104	Isha,Sharma	Delhi
5	P105	Nisha,Patel	Chennai
6	P107	Rahul,Singh	Chennai

/*9. Write a query to display profile id of the passenger(s) who has/have booked maximum number of tickets. In case of multiple records, display the records sorted in ascending order based on profile id.*/

```
SELECT app.Profile_ID
FROM air_passenger_profile app
JOIN air_ticket_info ati
ON ati.Profile_ID=app.Profile_ID
GROUP BY app.Profile_ID,app.First_Name,app.Address
HAVING COUNT(ati.TICKET_ID)=(
    SELECT MAX(ticket_count)
    FROM(
        SELECT COUNT(*) AS ticket_count
        FROM air_ticket_info
        GROUP BY PROFILE_ID
    )x
)
ORDER BY app.Profile_ID;
```

Output:

Results		Messages	
	Profile_ID		
1	P101		
2	P103		
3	P104		

/*10. Write a query to display the total number of tickets as “No_of_Tickets” booked in each flight in ABC Airlines. The Query should display the flight_id, from_location, to_location and the number of tickets. Display only the flights in which atleast 1 ticket is booked. Display the records sorted in ascending order based on flight id.*/

```
SELECT af.Flight_ID,af.From_Location,af.To_Location,COUNT(ati.Ticket_ID) AS  
No_Of_Tickets  
FROM air_flight af  
JOIN air_ticket_info ati  
ON af.Flight_ID=ati.Flight_ID  
GROUP BY af.Flight_ID,af.From_Location,af.To_Location  
HAVING COUNT(ati.Ticket_ID)>=1  
ORDER BY af.Flight_ID;
```

Output:

	Flight_ID	From_Location	To_Location	No_Of_Tickets
1	F101	Chennai	Hyderabad	3
2	F102	Hyderabad	Delhi	2
3	F103	Chennai	Hyderabad	2
4	F105	Delhi	Chennai	1
5	F201	Mumbai	Hyderabad	2
6	F202	Hyderabad	Chennai	1
7	F205	Chennai	Hyderabad	2

Tables Creation and Insertions:

```
CREATE DATABASE AirLines;  
USE AirLines;
```

```
CREATE TABLE air_passenger_profile (  
    PROFILE_ID VARCHAR(10) PRIMARY KEY,  
    PASSWORD VARCHAR(10),  
    FIRST_NAME VARCHAR(10),  
    LAST_NAME VARCHAR(10),  
    ADDRESS VARCHAR(100),  
    MOBILE_NUMBER BIGINT,  
    EMAIL_ID VARCHAR(30)  
);
```

```
CREATE TABLE air_flight (  
    FLIGHT_ID VARCHAR(10) PRIMARY KEY,  
    AIRLINE_ID VARCHAR(10),  
    AIRLINE_NAME VARCHAR(30),  
    FROM_LOCATION VARCHAR(20),  
    TO_LOCATION VARCHAR(20),  
    DEPARTURE_TIME TIME,  
    ARRIVAL_TIME TIME,  
    DURATION TIME,  
    TOTAL_SEATS INT  
);
```

```
CREATE TABLE air_flight_details (  
    FLIGHT_ID VARCHAR(10),  
    FLIGHT_DEPARTURE_DATE DATE,  
    PRICE DECIMAL(8,2),  
    AVAILABLE_SEATS INT,  
    PRIMARY KEY(FLIGHT_ID, FLIGHT_DEPARTURE_DATE),  
    FOREIGN KEY (FLIGHT_ID) REFERENCES air_flight(FLIGHT_ID)  
);
```

```
CREATE TABLE air_ticket_info (  
    TICKET_ID VARCHAR(10) PRIMARY KEY,  
    PROFILE_ID VARCHAR(10),  
    FLIGHT_ID VARCHAR(10),  
    FLIGHT_DEPARTURE_DATE DATE,  
    STATUS VARCHAR(10),  
    FOREIGN KEY (PROFILE_ID) REFERENCES air_passenger_profile(PROFILE_ID),  
    FOREIGN KEY (FLIGHT_ID, FLIGHT_DEPARTURE_DATE) REFERENCES  
air_flight_details(FLIGHT_ID, FLIGHT_DEPARTURE_DATE)  
);
```

```

CREATE TABLE air_credit_card_details (
    PROFILE_ID VARCHAR(10),
    CARD_NUMBER BIGINT,
    CARD_TYPE VARCHAR(10),
    EXPIRATION_MONTH INT,
    EXPIRATION_YEAR INT,
    FOREIGN KEY (PROFILE_ID) REFERENCES air_passenger_profile(PROFILE_ID)
);

INSERT INTO air_passenger_profile VALUES
('P101', 'pass1', 'Ravi', 'Kumar', 'Chennai', 9876543210, 'ravi@mail.com'),
('P102', 'pass2', 'Sita', 'Rao', 'Hyderabad', 9876543211, 'sita@mail.com'),
('P103', 'pass3', 'Arjun', 'Mehta', 'Bangalore', 9876543212, 'arjun@mail.com'),
('P104', 'pass4', 'Isha', 'Sharma', 'Delhi', 9876543213, 'isha@mail.com'),
('P105', 'pass5', 'Nisha', 'Patel', 'Chennai', 9876543214, 'nisha@mail.com'),
('P106', 'pass6', 'Karan', 'Verma', 'Mumbai', 9876543215, 'karan@mail.com'),
('P107', 'pass7', 'Rahul', 'Singh', 'Chennai', 9876543216, 'rahul@mail.com'),
('P108', 'pass8', 'Priya', 'Reddy', 'Hyderabad', 9876543217, 'priya@mail.com'),
('P109', 'pass9', 'Meera', 'Das', 'Kolkata', 9876543218, 'meera@mail.com'),
('P110', 'pass10', 'Vikram', 'Nair', 'Chennai', 9876543219, 'vikram@mail.com');

INSERT INTO air_flight VALUES
('F101', 'A001', 'ABC
Airlines', 'Chennai', 'Hyderabad', '08:00:00', '10:00:00', '02:00:00', 180),
('F102', 'A001', 'ABC
Airlines', 'Hyderabad', 'Delhi', '11:00:00', '13:30:00', '02:30:00', 180),
('F103', 'A001', 'ABC
Airlines', 'Chennai', 'Hyderabad', '18:00:00', '20:00:00', '02:00:00', 180),
('F104', 'A001', 'ABC
Airlines', 'Bangalore', 'Mumbai', '09:30:00', '11:30:00', '02:00:00', 170),
('F105', 'A001', 'ABC
Airlines', 'Delhi', 'Chennai', '06:00:00', '08:30:00', '02:30:00', 190),
('F201', 'A002', 'King
Airlines', 'Mumbai', 'Hyderabad', '10:00:00', '12:00:00', '02:00:00', 200),
('F202', 'A002', 'King
Airlines', 'Hyderabad', 'Chennai', '14:00:00', '16:00:00', '02:00:00', 190),
('F203', 'A003', 'SkyJet', 'Chennai', 'Delhi', '07:00:00', '09:30:00', '02:30:00', 185),
('F204', 'A003', 'SkyJet', 'Delhi', 'Mumbai', '21:00:00', '23:00:00', '02:00:00', 175),
('F205', 'A004', 'Indigo', 'Chennai', 'Hyderabad', '12:00:00', '14:00:00', '02:00:00', 200);

INSERT INTO air_flight_details VALUES
('F101', '2025-01-10', 4500, 120),
('F101', '2025-04-05', 5000, 110),
('F102', '2025-02-15', 7200, 140),
('F103', '2025-03-20', 4600, 130),
('F103', '2025-04-25', 5100, 125),
('F105', '2025-01-28', 6500, 150),
('F201', '2025-02-08', 6000, 170),
('F202', '2025-03-10', 5800, 160),
('F205', '2025-04-12', 5300, 175),
('F104', '2025-02-18', 7000, 160);

```

```
INSERT INTO air_ticket_info VALUES
('T001', 'P101', 'F101', '2025-01-10', 'BOOKED'),
('T002', 'P101', 'F101', '2025-04-05', 'BOOKED'),
('T003', 'P102', 'F103', '2025-03-20', 'BOOKED'),
('T004', 'P103', 'F102', '2025-02-15', 'BOOKED'),
('T005', 'P103', 'F105', '2025-01-28', 'BOOKED'),
('T006', 'P103', 'F103', '2025-04-25', 'BOOKED'),
('T007', 'P104', 'F201', '2025-02-08', 'BOOKED'),
('T008', 'P105', 'F205', '2025-04-12', 'BOOKED'),
('T009', 'P106', 'F202', '2025-03-10', 'BOOKED'),
('T010', 'P107', 'F101', '2025-01-10', 'BOOKED'),
('T011', 'P101', 'F102', '2025-02-15', 'BOOKED'),
('T012', 'P104', 'F201', '2025-02-08', 'BOOKED'),
('T013', 'P104', 'F205', '2025-04-12', 'BOOKED');
```

```
INSERT INTO air_credit_card_details VALUES
('P101', 4567123412341234, 'VISA', 11, 2027),
('P102', 4567555512341234, 'MASTER', 10, 2026),
('P103', 4567999912341234, 'VISA', 12, 2028),
('P104', 4567111112341234, 'RUPAY', 09, 2025),
('P105', 4567222212341234, 'VISA', 01, 2029),
('P106', 4567333312341234, 'MASTER', 02, 2030),
('P107', 4567444412341234, 'VISA', 05, 2028),
('P108', 4567555512345678, 'RUPAY', 03, 2027),
('P109', 4567666612342222, 'VISA', 04, 2031),
('P110', 4567777712349999, 'MASTER', 06, 2029);
```